

Jiyong Woo

其他

基本信息

姓名: Jiyong Woo

所属院校: 庆北国立大学

地区: South Korea

学科领域

材料物理

微电子科学与工程

集成电路设计与集成系统

研究领域

薄膜材料与表面物理

存储器设计与架构

微电子器件与制造

纳米电子与先进工艺

半导体器件物理与工艺

半导体与光电子材料

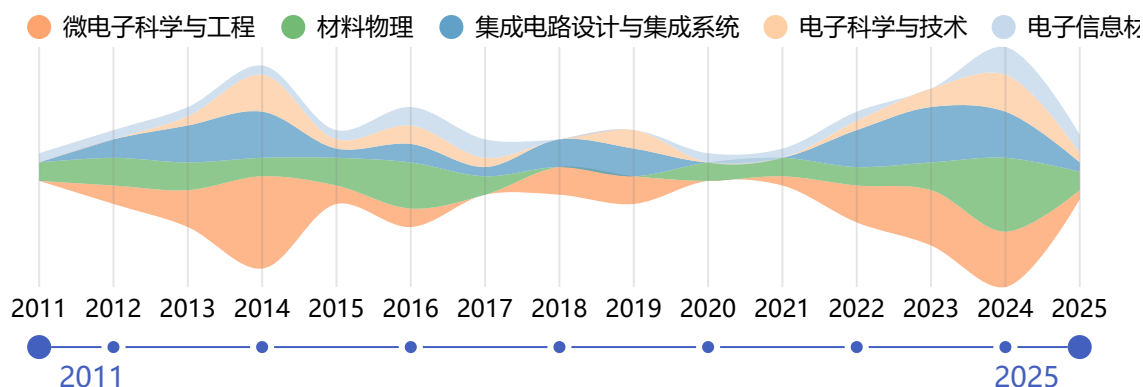
深度学习与神经网络

纳米物理与纳米器件

个人简介

Jiyong Woo is from Kyungpook National University. His research interests include semiconductor memory devices and neuromorphic systems. He has published numerous papers in relevant fields, such as "Neuromorphic computing using non - volatile memory" in ADVANCES IN PHYSICS: X in 2016, and "Improved Synaptic Behavior under Identical Pulses using AlOx/HfO2 Bilayer RRAM Array for Neuromorphic Systems" in IEEE Electron Device Letters in 2016. His works have received a significant number of citations, with some papers being cited over a hundred times.

近年发表的论文



Unraveling the role of post-annealing in IGZO transistor for memory applications

MICROELECTRONIC ENGINEERING

Kim, Nayeon; Jeong, Jiae; Lee, Jae Woo; Woo, Jiyong

被引用次数: 0

2025

Interface effect based nano-scale TiOX vertical synapse device for high-density integration in neuromorphic computing system

NANOTECHNOLOGY

Cho, Seojin; Han, Geonhui; Lee, Chuljun; Woo, Jiyong; Lee, Daeseok

被引用次数: 0

2025

Optimizing electrochemical and ferroelectric synaptic devices: from material selection to performance tuning

NEUROMORPHIC COMPUTING AND ENGINEERING

Kim, Eunjin; Jeon, Seonuk; Park, Hyoungjin; Jeong, Jiae; Choi, Hyeonsik; Kim, Yunsur; Kim, Jihyun; Lim, Seokjae; Moon, Kibong; Woo, Jiyong

被引用次数: 0

2025

Physics-Based SPICE-Compatible Compact Model of FLASH Memory With Poly-Si Channel for Computing-in-Memory Applications

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Cho, Jung Rae; Ryu, Donghyun; Kim, Donguk; Kim, Wonjung; Kim, Yeonwoo; Kim, Changwook; Kim, Yoon; Kang, Myounggon; Woo, Jiyong; Kim, Dae Hwan

被引用次数: 0

2025

Effect of the number and distribution of Al₂O₃ atomic layer deposition cycles within HfO₂ layer on ferroelectric characteristics

APPLIED PHYSICS LETTERS

Park, Hyoungjin; Jeong, Jiae; Kim, Hyun Wook; Hong, Eunryeong; Kim, Nayeon; Jeon, Seonuk; Kim, Yunsur; Choi, Hyeonsik; Woo, Jiyong

被引用次数: 0

2024

Understanding the role of GeSbTe layer on atomic switch through current sweep mode measurement for enhanced threshold switching device

AIP ADVANCES

Lim, Seokjae; Woo, Jiyong

Role of oxide barrier in a NbOx layer with noninert electrodes for high-yield threshold switching characteristics

AIP ADVANCES

Choi, Hyeonsik; Kim, Hyun Wook; Hong, Eunryeong; Kim, Nayeon; Jeon, Seonuk; Kim, Yunsur; Woo, Jiyong

被引用次数: 0

2024

Electrochemical random-access memory: recent advances in materials, devices, and systems towards neuromorphic computing

NANO CONVERGENCE

Kwak, Hyunjeong; Kim, Nayeon; Jeon, Seonuk; Kim, Seyoung; Woo, Jiyong

被引用次数: 4

2024

Effect of RF power on analog synaptic behavior of sputter-deposited InGaZnO films for neuromorphic computing applications

CERAMICS INTERNATIONAL

Shin, Dong Hyeop; Myoung, Seung Joo; Kim, Donguk; Kim, Changwook; Bae, Jong -Ho; Choi, Sung-Jin; Kim, Dong Myong; Woo, Jiyong; Kim, Dae Hwan

被引用次数: 0

2024

Internal Resistor Effect of Multilayer-Structured Synaptic Device for Low-Power Operation

NANOMATERIALS

Kim, Hyejin; Han, Geonhui; Cho, Seojin; Woo, Jiyong; Lee, Daeseok

被引用次数: 1

2024

Improvement of the Symmetry and Linearity of Synaptic Weight Update by Combining the InGaZnO Synaptic Transistor and Memristor

IEEE ACCESS

Yang, Tae Jun; Cho, Jung Rae; Lee, Hyunkyu; Lee, Hee Jun; Myoung, Seung Joo; Lee, Da Yeon; Choi, Sung-Jin; Bae, Jong-Ho; Kim, Dong Myong; Kim, Changwook; Woo, Jiyong; Kim, Dae Hwan

被引用次数: 1

2024

Hf_{0.4}Zr_{0.6}O₂ Thickness-Dependent Transfer Characteristics of In_xZn_{1-x}O_y Channel Ferroelectric FETs

JOURNAL OF PHYSICAL CHEMISTRY LETTERS

Jeong, Jiae; Park, Hyoungjin; Kim, Jihyun; Moon, Hojin; Choi, Hyeonsik; Kim, Eunjin; Jeon, Seonuk; Kim, Yunsur; Woo, Jiyong

被引用次数: 0

2024

Back-End-of-Line Compatible Transistor With Ultrathin InZnO Channel Enabling Operations in Cryogenic Environments

IEEE ELECTRON DEVICE LETTERS

Kim, Nayeon; Park, Hyoungjin; Jeong, Jiae; Kim, Hyun Wook; Hong, Eunryeong; Jeon, Seonuk; Choi, Hyeonsik; Kim, Yunsur; Woo, Jiyong

被引用次数: 1

2024

Linear Synaptic Weight Update in Selector-Less HfO₂ RRAM Using Al₂O₃ Built-In Resistor for Neuromorphic Computing Systems

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Yunsur; Kim, Hyejin; Jeon, Seonuk; Kim, Hyun Wook; Hong, Eunryeong; Kim, Nayeon; Choi, Hyeonsik; Park, Hyoungjin; Jeong, Jiae; Lee, Daeseok; Woo, Jiyong

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2024

Dual Ferroelectric Stack of HfZrO₂ /Al:HfO₂ With Tunable Coercive Voltage for High-Density Memory Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jeong, Jiae; Park, Hyoungjin; Kim, Nayeon; Kim, Hyun Wook; Hong, Eunryeong; Choi, Hyeonsik; Jeon, Seonuk; Kim, Yunsur; Woo, Jiyong

被引用次数: 0

2024

Demonstration of Threshold Switching in Undoped SiOX Layer for Oscillation Neurons

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hong, Eunryeong; Kim, Hyun Wook; Choi, Hyeonsik; Jeon, Seonuk; Kim, Nayeon; Woo, Jiyong

被引用次数: 0

2024

Retention-aware zero-shifting technique for Tiki-Taka algorithm-based analog deep learning accelerator

SCIENCE ADVANCES

Noh, Kyungmi; Kwak, Hyunjeong; Son, Jeonghoon; Kim, Seungkun; Um, Minseong; Kang, Minil; Kim, Doyoon; Ji, Wonjae; Lee, Junyong; Jo, Hwijee; Woo, Jiyong; Lee, Hyung-Min; Kim, Seyoung

被引用次数: 0

2024

Strategy to improve synaptic behavior of ion-actuated synaptic transistors-the use of ion blocking layer to improve state retention

SCIENTIFIC REPORTS

Jeon, Seonuk; Tessler, Nir; Kim, Nayeon; Hong, Eunryeong; Kim, Hyun Wook; Woo, Jiyong

被引用次数: 0

2024

Controllable Polarization Switching of Hafnia-Based Ferroelectric Bilayers

2024 IEEE SILICON NANOELECTRONICS WORKSHOP, SNW 2024

Jeong, Jiae; Park, Hyoungjin; Woo, Jiyong

被引用次数: 0

2024

Highly Controllable Probabilistic Bits Leveraging Instability of SiO_x-Based Threshold Switching Devices for Probabilistic Computing

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Choi, Hyeonsik; Kim, Hyun Wook; Hong, Eunryeong; Kim, Nayeon; Jeon, Seonuk; Kim, Yunsur; Park, Hyoungjin; Jeong, Jiae; Woo, Jiyong

被引用次数: 0

2024

Engineering Strategies in HfO_x RRAM-Based Analog Synapses Toward Linear Weight Update for Neuromorphic Hardware Accelerators

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Kim, Yunsur; Choi, Hyeonsik; Woo, Jiyong

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2024

Switching mechanisms of CMOS-compatible ECRAM transistors-Electrolyte charging and ion plating

JOURNAL OF APPLIED PHYSICS

Tessler, Nir; Kim, Nayeon; Kang, Heebum; Woo, Jiyong

被引用次数: 5

2023

Understanding ovonic threshold switching of GeSe chalcogenide materials using electrical methodologies for extracting density of states

APPLIED PHYSICS LETTERS

Kim, Donguk; Yang, Tae Jun; Choi, Woo Sik; Lee, Hee Jun; Jang, Jun Tae; Hong, Eunryeong; Yang, Woo Young; Choi, Minwoo; Yang, Ki Yeon; Lee, Chang Seung; Woo, Jiyong; Kim, Dae Hwan

被引用次数: 0

2023

Interrelation between ferroelectric properties and defects based on low-frequency noise analysis of HZO ferroelectric capacitor

APPLIED PHYSICS LETTERS

Jin, Seunghee; Jeon, Juye; Kim, Min Jung; Heo, Kiseok; Kim, Jeong Hun; Im, Jong-Pil; Yoon, Sung-Min; Moon, Seung Eon; Woo, Jiyong; Lee, Jae Woo

被引用次数: 4

2023

Convolutional kernel with PrCaMnOx-based resistive random-access memory for neuromorphic image processing

AIP ADVANCES

Hong, Eunryeong; Jeon, Seonuk; Kim, Nayeon; Kim, Hyun Wook; Kang, Heebum; Moon, Kibong; Woo, Jiyong

被引用次数: 2

2023

Definition of a Localized Conducting Path via Suppressed Charge Injection in Oxide Memristors for Stable Practical Hardware Neural Networks

ACS APPLIED MATERIALS & INTERFACES

Kim, Hyeongwook; Lee, Jihwan; Kim, Hyun Wook; Woo, Jiyong; Kim, Min-Hwi; Lee, Sin-Hyung

被引用次数: 3

2023

Brain-Inspired Mutual Synchronization in Cross-Coupled NbOx Oscillation Neurons for Oscillatory Neural Network Applications

IEEE ACCESS

Kim, Hyun Wook; Jeon, Seyeong; Jeon, Seonuk; Hong, Eunryeong; Kim, Nayeon; Woo, Jiyong

被引用次数: 1

2023

Experimental Robust Spontaneous Synchronizations in Coupled NbOx Oscillation Neurons for Unconventional Computing

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Hyun Wook; Kang, Seung-Youl; Moon, Jaehyun; Kim, Nayeon; Hong, Eunryeong; Jeon, Seyeong; Jeon, Seonuk; Woo, Jiyong

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2023

Understanding Rhythmic Synchronization of Oscillatory Neural Networks Based on NbOx Artificial Neurons for Edge Detection

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Hyun Wook; Jeon, Seonuk; Kang, Heebum; Hong, Eunryeong; Kim, Nayeon; Woo, Jiyong

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2023

Hybrid Precision in Resistive Memory-Based Convolutional Kernel for Fault-Resilient Neuromorphic Systems

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jeon, Seonuk; Hong, Eunryeong; Kang, Heebum; Kim, Hyun Wook; Kim, Nayeon; Woo, Jiyong

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2023

Read Disturbances in Cross-Point Phase-Change Memory Arrays- Part I: Physical Modeling With Phase-Change Dynamics

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Donguk; Jang, Jun Tae; Kim, Changwook; Kim, Hyun Wook; Hong, Eunryeong; Ban, Sanghyun; Shin, Minchul; Lee, Hanwool; Lee, Hyung Dong; Mo, Hyun-Sun; Woo, Jiyong; Kim, Dae Hwan

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2023

Read Disturbance in Cross-Point Phase-Change Memory Arrays-Part II: Array Simulations Considering External Currents

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Donguk; Jang, Jun Tae; Kim, Changwook; Kim, Hyun Wook; Hong, Eunryeong; Ban, Sanghyun; Shin, Minchul; Lee, Hanwool; Lee, Hyung Dong; Mo, Hyun-Sun; Woo, Jiyong; Kim, Dae Hwan

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2023

WOx channel engineering of Cu-ion-driven synaptic transistor array for low-power neuromorphic computing

SCIENTIFIC REPORTS

Jeon, Seonuk; Kang, Heebum; Kwak, Hyunjeong; Noh, Kyungmi; Kim, Seungkun; Kim, Nayeon; Kim, Hyun Wook; Hong, Eunryeong; Kim, Seyoung; Woo, Jiyong

被引用次数: 2

2023

Three-dimensional vertical structural electrochemical random access memory for high-density integrated synapse device

SCIENTIFIC REPORTS

Kim, Hyejin; Seo, Jongseon; Cho, Seojin; Jeon, Seonuk; Woo, Jiyong; Lee, Daeseok

被引用次数: 4

2023

Understanding synaptic characteristics of nonvolatile analog redox transistor based on mobile ion-modulated-electrolyte thickness model for neuromorphic applications

APPLIED PHYSICS LETTERS

Kim, Nayeon; Kang, Heebum; Kim, Hyun Wook; Hong, Eunryeong; Woo, Jiyong

被引用次数: 7

2022

Analog synaptic behavior of mobile ion source-limited electrochemical RAM using CuOx oxide electrode for deep learning accelerator

APPLIED PHYSICS LETTERS

Kang, Heebum; Kim, Hyun Wook; Hong, Eun Ryeong; Woo, Jiyong

被引用次数: 9

2022

Improvement in nonvolatile memory operations for metal-ferroelectric-insulator-semiconductor capacitors using HfZrO₂ and ZrO₂ thin films as ferroelectric and insulator layers

JOURNAL OF PHYSICS D-APPLIED PHYSICS

Kim, Yeriaron; Kang, Seung Youl; Woo, Jiyong; Kim, Jeong Hun; Im, Jong-Pil; Yoon, Sung-Min; Moon, Seung Eon

被引用次数: 1

2022

Exploiting Read Current Noise of TiOx Resistive Memory by Controlling Forming Conditions for Probabilistic Neural Network Hardware

IEEE ELECTRON DEVICE LETTERS

Choi, Wooseok; Ji, Wonjae; Heo, Seongjae; Lee, Donguk; Noh, Kyungmi; Lee, Chuljun; Woo, Jiyong; Kim, Seyoung; Hwang, Hyunsang

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2022

Area and Thickness Scaling of NbOx-Based Threshold Switches for Oscillation Neurons

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Kim, Hyun Wook; Kang, Heebum; Hong, Eunryeong; Woo, Jiyong

被引用次数: 7

2022

Ion-Driven Electrochemical Random-Access Memory-Based Synaptic Devices for Neuromorphic Computing Systems: A Mini-Review

MICROMACHINES

Kang, Heebum; Seo, Jongseon; Kim, Hyejin; Kim, Hyun Wook; Hong, Eun Ryeong; Kim, Nayeon; Lee, Daeseok; Woo, Jiyong

被引用次数: 9

2022

Linear Frequency Modulation of NbO₂-Based Nanoscale Oscillator With Li-Based Electrochemical Random Access Memory for Compact Coupled Oscillatory Neural Network

FRONTIERS IN NEUROSCIENCE

Lee, Donguk; Kwak, Myonghoon; Lee, Jongwon; Woo, Jiyong; Hwang, Hyunsang

被引用次数: 1

2022

Analysis of electro-chemical RAM synaptic array for energy-efficient weight update

FRONTIERS IN NANOTECHNOLOGY

Kang, Heebum; Kim, Nayeon; Jeon, Seonuk; Kim, Hyun Wook; Hong, Eunryeong; Kim, Seyoung; Woo, Jiyong

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2022

Hybrid memory characteristics of NbOx threshold switching devices

APPLIED PHYSICS LETTERS

Lee, Sangmin; Hwang, Hyunsang; Woo, Jiyong

被引用次数: 6

2021

Cu-ion-actuated three-terminal neuromorphic synaptic devices based on binary metal-oxide electrolyte and channel

APPLIED PHYSICS LETTERS

Kang, Heebum; Woo, Jiyong

被引用次数: 16

2021

Two- and three-terminal HfO₂-based multilevel resistive memories for neuromorphic analog synaptic elements

NEUROMORPHIC COMPUTING AND ENGINEERING

Kang, Heebum; Park, Jinah; Lee, Dokyung; Kim, Hyun Wook; Jin, Sol; Ahn, Minjoon; Woo, Jiyong

被引用次数: 12

2021

Optimized annealing conditions to enhance stability of polarization in sputtered HfZrO_x layers for non-volatile memory applications

CURRENT APPLIED PHYSICS

Kim, Yeriaron; Woo, Jiyong; Im, Solyee; Lee, Yeseul; Kim, Jeong Hun; Im, Jong-Pil; Suh, Dongwoo; Yang, Sang Mo; Yoon, Sung-Min; Moon, Seung Eon

被引用次数: 12

2020

Artificial Perspiration Membrane by Programmed Deformation of Thermoresponsive Hydrogels

ADVANCED MATERIALS

Kim, Junsoo; Im, Solyee; Kim, Jeong Hun; Kim, Sang Moon; Lee, Seung-Min; Lee, Jaewoo; Im, Jong Pil; Woo, Jiyong; Moon, Seung Eon

被引用次数: 26

2020

Recent Advancements in Emerging Neuromorphic Device Technologies

ADVANCED INTELLIGENT SYSTEMS

Woo, Jiyong; Kim, Jeong Hun; Im, Jong-Pil; Moon, Seung Eon

被引用次数: 20

2020

Ferroelectric Switching in Trilayer Al₂O₃/HfZrO_x/Al₂O₃ Structure

MICROMACHINES

Im, Solyee; Kang, Seung-Youl; Kim, Yeriaron; Kim, Jeong Hun; Im, Jong-Pil; Yoon, Sung-Min; Moon, Seung Eon; Woo, Jiyong

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2020

Microstructural engineering in interface-type synapse device for enhancing linear and symmetric conductance changes

NANOTECHNOLOGY

Park, Jaesung; Lee, Chuljun; Kwak, Myunghoon; Chekol, Solomon Amsalu; Lim, Seokjae; Kim, Myungjun; Woo, Jiyong; Hwang, Hyunsang; Lee, Daeseok

被引用次数: 20

2019

RRAM-based synapse devices for neuromorphic systems

FARADAY DISCUSSIONS

Moon, K.; Lim, S.; Park, J.; Sung, C.; Oh, S.; Woo, J.; Lee, J.; Hwang, H.

被引用次数: 181

2019

Energy-Storing Hybrid 3D Vertical Memory Structure

IEEE ELECTRON DEVICE LETTERS

Kim, Myungjun; Lee, Chuljun; Song, Yubin; Koo, Sang-Mo; Oh, Jong-Min; Woo, Jiyong; Lee, Daeseok

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2019

Integrated Crossbar Array With Resistive Synapses and Oscillation Neurons

IEEE ELECTRON DEVICE LETTERS

Woo, Jiyong; Wang, Panni; Yu, Shimeng

被引用次数: 31

2019

Two-Terminal Structured Synaptic Device Using Ionic Electrochemical Reaction Mechanism for Neuromorphic System

IEEE ELECTRON DEVICE LETTERS

Lee, Chuljun; Lee, Jaewon; Kim, Myungjun; Woo, Jiyong; Koo, Sang-Mo; Oh, Jong-Min; Lee, Daeseok

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2019

Impact of Selector Devices in Analog RRAM-Based Crossbar Arrays for Inference and Training of Neuromorphic System

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Woo, Jiyong; Yu, Shimeng

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2019

CMOS compatible low-power volatile atomic switch for steep-slope FET devices

APPLIED PHYSICS LETTERS

Lim, Seokjae; Yoo, Jongmyung; Song, Jeonghwan; Woo, Jiyong; Park, Jaehyuk; Hwang, Hyunsang

被引用次数: 9

2018

Comparative Study of Cross-Point MRAM Array With Exponential and Threshold Selectors for Read Operation

IEEE ELECTRON DEVICE LETTERS

Woo, Jiyong; Yu, Shimeng

被引用次数: 14

2018

Two-Step Read Scheme in One-Selector and One-RRAM Crossbar-Based Neural Network for Improved Inference Robustness

IEEE TRANSACTIONS ON ELECTRON DEVICES

Woo, Jiyong; Yu, Shimeng

Controllable quantized conductance for multilevel data storage applications using conductive bridge random access memory

NANOTECHNOLOGY

Aga, Fekadu Gochole; Woo, Jiyong; Song, Jeonghwan; Park, Jaehyuk; Lim, Seokjae; Sung, Changhyuck; Hwang, Hyunsang

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2017

Self-Limited CBRAM With Threshold Selector for 1S1R Crossbar Array Applications

IEEE ELECTRON DEVICE LETTERS

Song, Jeonghwan; Woo, Jiyong; Lim, Seokjae; Chekol, Solomon Amsalu; Hwang, Hyunsang

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2017

Linking Conductive Filament Properties and Evolution to Synaptic Behavior of RRAM Devices for Neuromorphic Applications

IEEE ELECTRON DEVICE LETTERS

Woo, Jiyong; Padovani, Andrea; Moon, Kibong; Kwak, Myounghun; Larcher, Luca; Hwang, Hyunsang

被引用次数: 75

2017

Comprehensive scaling study of NbO₂ insulator-metal-transition selector for cross point array application

APPLIED PHYSICS LETTERS

Cha, Euijun; Park, Jaehyuk; Woo, Jiyong; Lee, Daeseok; Prakash, Amit; Hwang, Hyunsang

被引用次数: 99

2016

Improved reset breakdown strength in a HfO_x-based resistive memory by introducing RuO_x oxygen diffusion barrier

AIP ADVANCES

Park, Jaesung; Woo, Jiyong; Prakash, Amit; Lee, Sangheon; Lim, Seokjae; Hwang, Hyunsang

被引用次数: 2

2016

Retention modeling for ultra-thin density of Cu-based conductive bridge random access memory (CBRAM)

AIP ADVANCES

Aga, Fekadu Gochole; Woo, Jiyong; Lee, Sangheon; Song, Jeonghwan; Park, Jaesung; Park, Jaehyuk; Lim, Seokjae; Sung, Changhyuck; Hwang, Hyunsang

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2016

Improved Synaptic Behavior Under Identical Pulses Using AlO_x/HfO₂ Bilayer RRAM Array for Neuromorphic Systems

IEEE ELECTRON DEVICE LETTERS

Woo, Jiyong; Moon, Kibong; Song, Jeonghwan; Lee, Sangheon; Kwak, Myounghun; Park, Jaesung; Hwang, Hyunsang

被引用次数: 414

2016

Steep Slope Field-Effect Transistors With Ag/TiO₂-Based Threshold Switching Device

IEEE ELECTRON DEVICE LETTERS

Song, Jeonghwan; Woo, Jiyong; Lee, Sangheon; Prakash, Amit; Yoo, Jongmyung; Moon, Kibong; Hwang, Hyunsang

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2016

Role of Local Chemical Potential of Cu on Data Retention Properties of Cu-Based Conductive-Bridge RAM

IEEE ELECTRON DEVICE LETTERS

Woo, Jiyong; Belmonte, Attilio; Redolfi, Augusto; Hwang, Hyunsang; Jurczak, Malgorzata; Goux, Ludovic

被引用次数: 18

2016

Optimized Programming Scheme Enabling Linear Potentiation in Filamentary HfO₂ RRAM Synapse for Neuromorphic Systems

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Woo, Jiyong; Moon, Kibong; Song, Jeonghwan; Kwak, Myounghoon; Park, Jaesung; Hwang, Hyunsang

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2016

Introduction of WO₃ Layer in a Cu-Based Al₂O₃ Conductive Bridge RAM System for Robust Cycling and Large Memory Window

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Woo, Jiyong; Belmonte, Attilio; Redolfi, Augusto; Hwang, Hyunsang; Jurczak, Malgorzata; Goux, Ludovic

被引用次数: 29

2016

Effect of AC pulse overshoot on nonlinearity and reliability of selectorless resistive random access memory in AC pulse operation

SOLID-STATE ELECTRONICS

Lee, Sangheon; Song, Jeonghwan; Lee, Daeseok; Woo, Jiyong; Cha, Euijun; Hwang, Hyunsang

被引用次数: 9

2015

Threshold switching behavior of Ag-Si based selector device and hydrogen doping effect on its characteristics

AIP ADVANCES

Yoo, Jongmyung; Woo, Jiyong; Song, Jeonghwan; Hwang, Hyunsang

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2015

Threshold Selector With High Selectivity and Steep Slope for Cross-Point Memory Array

IEEE ELECTRON DEVICE LETTERS

Song, Jeonghwan; Woo, Jiyong; Prakash, Amit; Lee, Daeseok; Hwang, Hyunsang

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2015

Accelerated Retention Test Method by Controlling Ion Migration Barrier of Resistive Random Access Memory

IEEE ELECTRON DEVICE LETTERS

Koo, Yunmo; Ambrogio, Stefano; Woo, Jiyong; Song, Jeonghwan; Ielmini, Daniele; Hwang, Hyunsang

被引用次数: 11

2015

The band-gap energy dependence of metal oxides on non-linear characteristics in the HfO₂-based resistive random access memory

MICROELECTRONIC ENGINEERING

Lee, Sangheon; Lee, Daeseok; Woo, Jiyong; Cha, Euijun; Park, Jaesung; Moon, Kibong; Song, Jeonghwan; Hwang, Hyunsang

被引用次数: 6

2015

Dependence of reactive metal layer on resistive switching in a bi-layer structure Ta/HfO_x filament type resistive random access memory

APPLIED PHYSICS LETTERS

Lee, Daeseok; Woo, Jiyong; Park, Sangsu; Cha, Euijun; Lee, Sangheon; Hwang, Hyunsang

被引用次数: 21

2014

Tunnel barrier engineering of titanium oxide for high non-linearity of selector-less resistive random access memory

APPLIED PHYSICS LETTERS

Lee, Sangheon; Woo, Jiyong; Lee, Daeseok; Cha, Euijun; Park, Jaesung; Moon, Kibong; Song, Jeonghwan; Koo, Yunmo; Hwang, Hyunsang

被引用次数: 15

2014

Stepwise set operation for reliable switching uniformity and low operating current of ReRAMs

SOLID-STATE ELECTRONICS

Lee, Sangheon; Woo, Jiyong; Lee, Daeseok; Cha, Euijun; Hwang, Fiyunsang

被引用次数: 2

2014

Low-temperature spin-on-glass method involving high-pressure annealing for filling high-aspect-ratio structures

JAPANESE JOURNAL OF APPLIED PHYSICS

Lee, Daeseok; Woo, Jiyong; Park, Sangsu; Cha, Euijun; Lee, Sangheon; Hwang, Hyunsang

被引用次数: 1

2014

Internal resistor of multi-functional tunnel barrier for selectivity and switching uniformity in resistive random access memory

NANOSCALE RESEARCH LETTERS

Lee, Sangheon; Woo, Jiyong; Lee, Daeseok; Cha, Euijun; Hwang, Hyunsang

被引用次数: 16

2014

Hardware implementation of associative memory characteristics with analogue-type resistive-switching device

NANOTECHNOLOGY

Moon, Kibong; Park, Sangsu; Jang, Junwoo; Lee, Daeseok; Woo, Jiyong; Cha, Euijun; Lee, Sangheon; Park, Jaesung; Song, Jeonghwan; Koo, Yunmo; Hwang, Hyunsang

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2014

Engineering Oxygen Vacancy of Tunnel Barrier and Switching Layer for Both Selectivity and Reliability of Selector-Less ReRAM

IEEE ELECTRON DEVICE LETTERS

Lee, Sangheon; Lee, Daeseok; Woo, Jiyong; Cha, Euijun; Park, Jaesung; Hwang, Hyunsang

被引用次数: 21

2014

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IEEE ELECTRON DEVICE LETTERS

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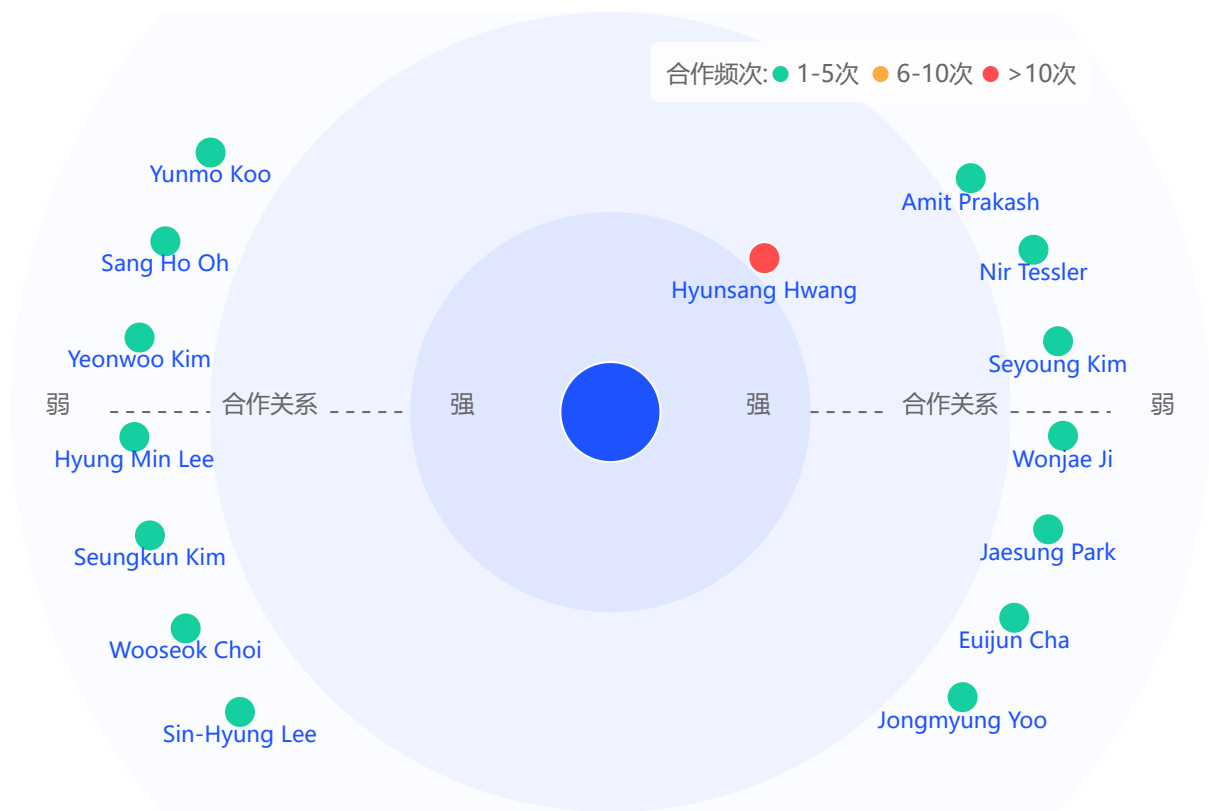
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